



Prediction of Medical Care Abandonment in Italy Using Population Survey Data

Machine Learning and Conformal Prediction on EHIS 2019

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Motivation and Objective

1 Motivation and Objective

- Forgoing necessary care leads to worse health outcomes and higher long-term costs.
- The 2019 EHIS offers pre-pandemic insights into Italian health and access to care.
- Our goal: classify individuals at risk of medical care abandonment.
- We model behavioral, demographic, and socio-health determinants.
- Example: middle-aged caregiver may delay care due to family duties and transport barriers.
- Aim: support fairer, data-driven health interventions.



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Survey Design

2 Data and Sampling Design

- Dataset: European Health Interview Survey (EHIS) 2019.
- 45,930 observations, 244 variables.
- Multistage stratified sample: 24,000 households across 797 municipalities.
- Statistically representative of the Italian population.
- Only direct responses were retained (proxy responses excluded, 2,800).
- Outcome: self-reported renunciation of care with 3 categories:
 - Class 1: Gave up care
 - Class 2: Needed care but did not renounce
 - Class 3: Had no need for care



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3 Exploratory Data Analysis

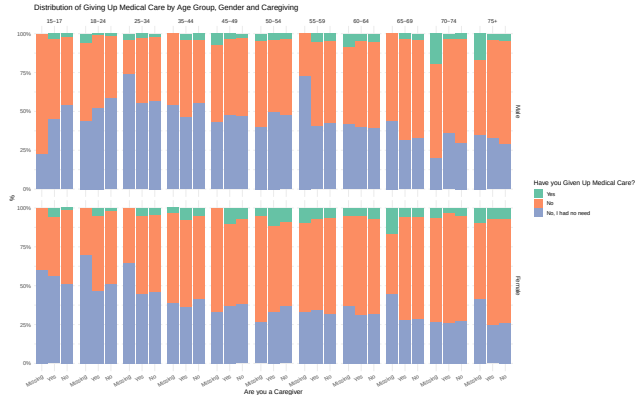
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Age, Gender, and Caregiving

3 Exploratory Data Analysis

- Probability of abandoning care increases with age.
- Women aged 40–60 (especially caregivers) are most affected.
- These "sandwich generation" cases face double caregiving burdens.

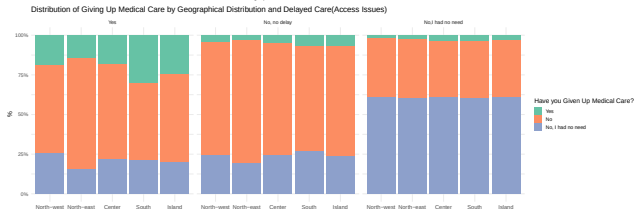
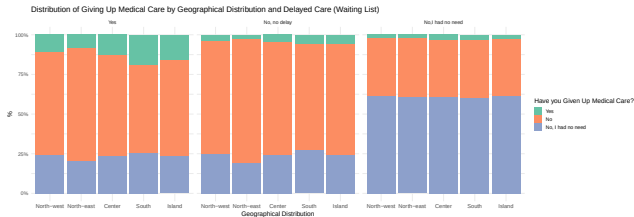




Delays and Geographical Disparities

3 Exploratory Data Analysis

- Delays due to waiting lists or transport barriers raise abandonment.
- Southern Italy and Islands show highest unmet needs.
- Confirms structural inequalities in healthcare access.





Perceived Social Attention

3 Exploratory Data Analysis

- Individuals reporting low social attention are more likely to forgo care.
- Community support seems protective: stronger ties → better care access.

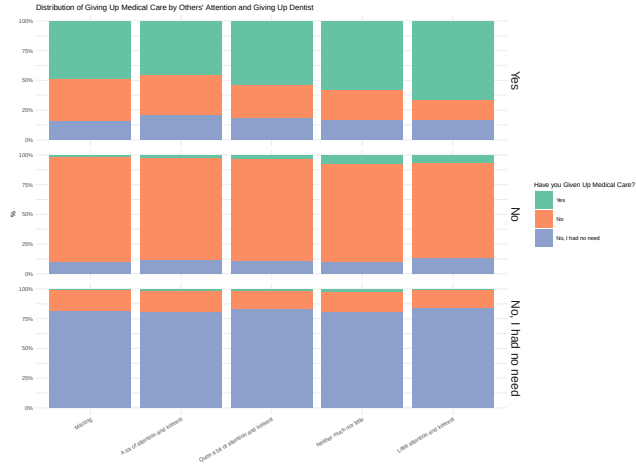




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Data Preprocessing

4 Preprocessing and Feature Engineering

- Grouped variables using regex into thematic domains.
- Managed skip logic and differentiated true vs logical missing values.
- Excluded proxy responses to ensure consistency.
- Final working dataset: 400 selected features (filtered from 244 variables).

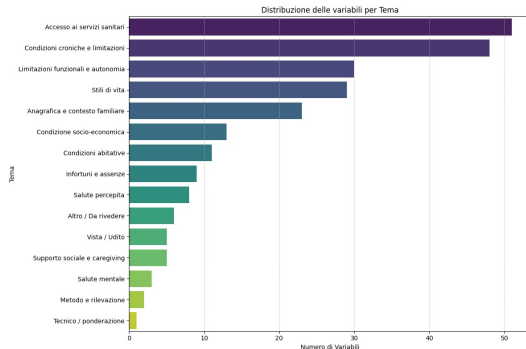




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Predictive Modeling Objective

5 Modelling

- Goal: Predict which individuals are likely to avoid or renounce medical care.
- Outcome: Assign each person to a specific healthcare behavior class.
- Two approaches: Multinomial Logistic Regression and XGBoost.



Multinomial Logistic Regression

5 Modelling

- Extension of logistic regression for multiple categories.
- Estimates the probability of each healthcare behavior class based on predictors (e.g., income, age, access barriers).
- Key advantages:
 - Directly interpretable effects for each variable.
 - Odds ratios show how predictors influence behavior.
- Uses LASSO regularization for feature selection.



XGBoost Classifier

5 Modelling

- Powerful machine learning method using decision tree ensembles.
- Learns complex, nonlinear relationships and interactions.
- Regularization helps prevent overfitting and improves generalization.
- Produces probability estimates for each class.
- Feature importance and interpretability via SHAP.



Why Use Both Models?

5 Modelling

- **Multinomial Logistic Regression:**
 - Highly interpretable, quantifies impact of each predictor.
- **XGBoost:**
 - Captures complex interactions, usually higher predictive accuracy.
- Comparing both gives robust, actionable, and explainable results.



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Conformal Prediction: Motivation & Advantages

6 Conformal Predictions

- Traditional metrics give a single “best guess” for each case.
- Conformal prediction produces a set of likely outcomes for each individual, with statistical guarantees.
- Key advantages:
 - Robust quantification of uncertainty
 - Few assumptions about the underlying data
 - Valid even when the data distribution shifts
 - Guarantees coverage in small samples



Mondrian Inductive Conformal Prediction

6 Conformal Predictions

- Improves on standard conformal prediction by calibrating uncertainty within each class.
- Especially effective for imbalanced data (rare behavior classes).
- Key steps:
 1. Train model, then calculate class-specific nonconformity scores on a calibration set.
 2. For each class, determine a threshold to guarantee desired coverage.
 3. For each test case, return all classes whose score is below its class threshold.



Why Bootstrap?

6 Conformal Predictions

- Single train-test splits are sensitive to random selection—results may vary by random seed.
- We use bootstrap resampling (100 iterations) to ensure stability and reproducibility.
- This approach evaluates feature selection and model performance across many different data samples.



Bootstrap Workflow

6 Conformal Predictions

1. Draw a bootstrap sample from the original dataset.
 2. Split sample: 60% training, 20% calibration, 20% test.
 3. Select features and fit models on training data.
 4. Evaluate performance and record which features are selected.
 5. Repeat 100 times for robust stability analysis.
- Outcome: Measures how consistently each predictor is identified as important.



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7 Feature Selection

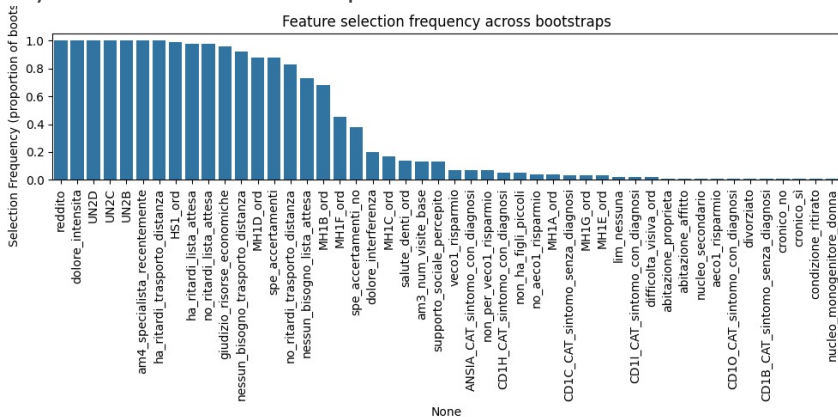
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Feature Selection Stability

7 Feature Selection

- LASSO (Logistic): Shrinks coefficients, selects most relevant predictors.
- Stability checked via 100 bootstraps.

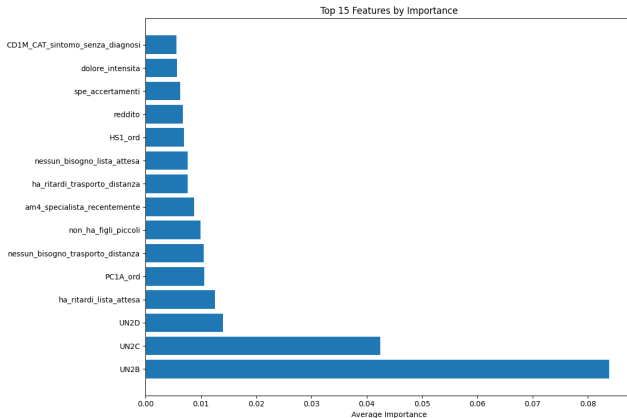




Key Predictors Identified

7 Feature Selection

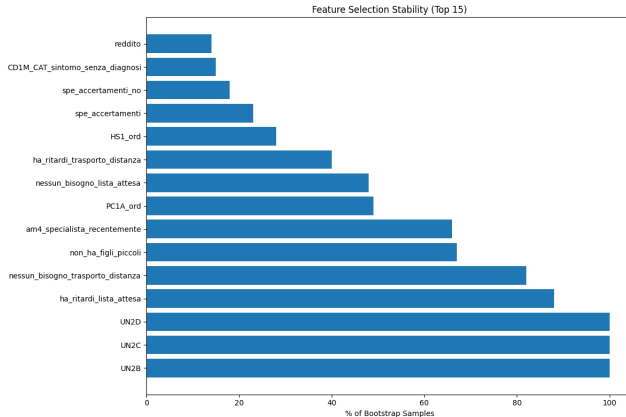
- Consistent top features:
 - Unmet needs (UN2B, UN2C, UN2D)
 - Income, age, access barriers





Feature Stability (XGBoost)

7 Feature Selection



- Top predictors stable across resamples.



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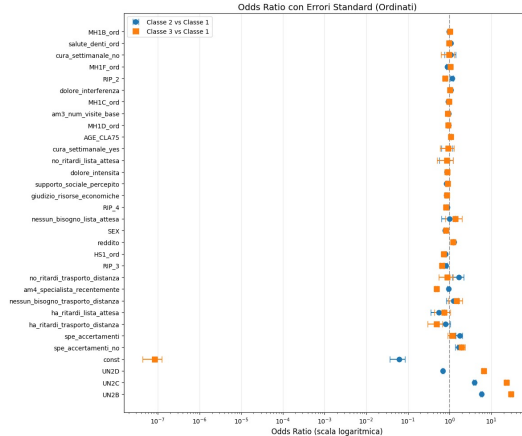
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Model Interpretation: Logistic Regression

8 Interpretability

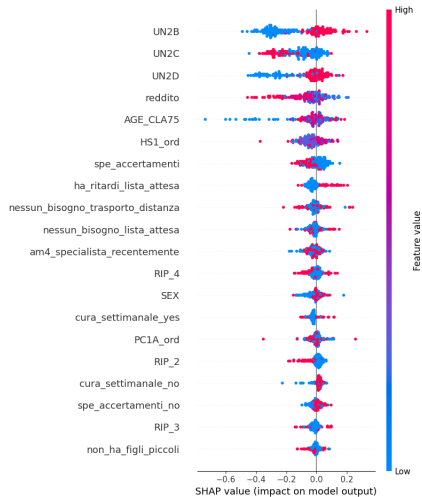


● Odds ratios show strong effects for unmet needs and access.



Model Interpretation: XGBoost + SHAP

8 Interpretability





SHAP Dependence Plot: Example

8 Interpretability

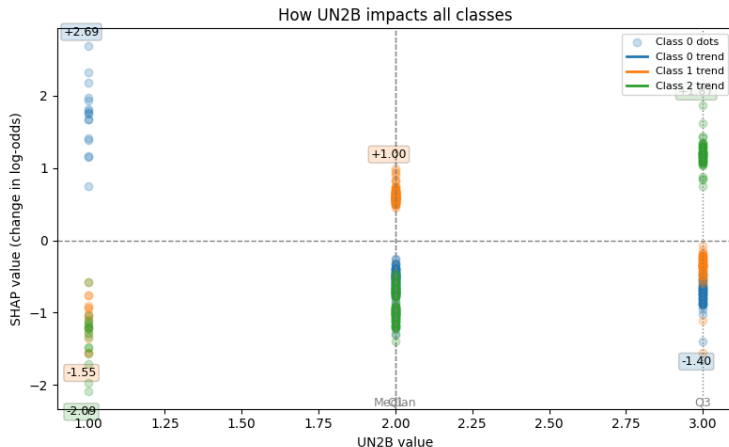




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Model Evaluation Metrics

9 Evaluation

- Precision, Recall, F1-Score: per class and overall.
- Population-weighted and unweighted performance.
- Confusion matrices: misclassification patterns.



Unweighted Confusion Matrix: Logistic Regression

9 Evaluation

	Pred 1	Pred 2	Pred 3	Total
True 1	333	72	38	443
True 2	853	3,429	518	4,800
True 3	131	329	2,984	3,444
Total	1,317	3,830	3,540	8,687

- Many false positives for Class 1.



Unweighted Confusion Matrix: XGBoost

9 Evaluation

	Pred 1	Pred 2	Pred 3	Total
True 1	341	68	31	440
True 2	227	4,148	424	4,799
True 3	77	338	3,028	3,443
Total	645	4,554	3,483	8,682

- Fewer false positives for Class 1 compared to Logistic Regression.



Unweighted Classification Metrics: Logistic Regression

9 Evaluation

Class	Precision	Recall	F1-score
1	0.25	0.73	0.37
2	0.89	0.72	0.80
3	0.85	0.86	0.85

- Low precision for Class 1 (many false alarms).



Unweighted Classification Metrics: XGBoost

9 Evaluation

Class	Precision	Recall	F1-score
1	0.47	0.76	0.58
2	0.90	0.85	0.88
3	0.87	0.87	0.87

- Substantially higher precision and F1 for Class 1.



Model Comparison: Unweighted Test Set

9 Evaluation

- XGBoost improves Class 1 F1-score by 57% (0.58 vs 0.37).
- Precision nearly doubles (0.47 vs 0.25).
- Fewer false positives for healthcare avoiders.
- Improved performance across all classes.



Population-Weighted Confusion Matrix: Logistic Regression

9 Evaluation

	Pred 1	Pred 2	Pred 3	Total
True 1	384,680	81,915	47,033	513,628
True 2	927,268	3,798,555	582,830	5,308,653
True 3	159,681	368,377	3,436,015	3,964,073
Total	1,471,629	4,248,847	4,065,878	9,786,354

- Many false positives for Class 1.



Population-Weighted Confusion Matrix: XGBoost

9 Evaluation

	Pred 1	Pred 2	Pred 3	Total
True 1	386,717	68,997	26,990	482,704
True 2	282,244	4,532,658	481,242	5,296,144
True 3	80,704	368,737	3,515,305	3,964,746
Total	749,665	4,970,392	4,023,537	9,743,594

- Major reduction in false positives for Class 1.



Population-Weighted Metrics: Logistic Regression

9 Evaluation

Class	Precision	Recall	F1-score
1	0.26	0.75	0.39
2	0.89	0.72	0.80
3	0.85	0.87	0.86

- Low F1-score for Class 1 due to many false positives.



Population-Weighted Metrics: XGBoost

9 Evaluation

Class	Precision	Recall	F1-score
1	0.52	0.80	0.63
2	0.91	0.86	0.88
3	0.87	0.89	0.88

- F1-score for Class 1 increases by 62% over Logistic Regression.



Conformal Prediction: Outputs & Evaluation

9 Evaluation

- Each case receives a prediction set (may contain more than one class).
- Two main metrics:
 - **Empirical coverage:** Fraction of test cases where the true class is in the set (should be at least 95%).
 - **Average set size:** How many classes are typically included (smaller = more informative).
- This framework complements accuracy with reliable uncertainty quantification—critical for policy.



Conformal Prediction Results

9 Evaluation

Model	Mean coverage	Mean set size
XGBoost	0.949 ± 0.004	1.29 ± 0.02
Logistic Regression	0.950 ± 0.003	1.86 ± 0.05

- Both models achieve target coverage; XGBoost more selective.